PLC STM32F407VET6 + W5500 : Modbus TCP (LAN), nap code qua USB-C (CH340C)

Nguon 24VDC

File: power.kicad_sch

MCU STM32F407VET6

File: mcu.kicad_sch

USB nap code

File: usb.kicad_sch

Ethernet

File: comm.kicad_sch

16 ngo ra transistor

File: outputs.kicad_sch

10 ngo vao analog

File: analog.kicad_sch

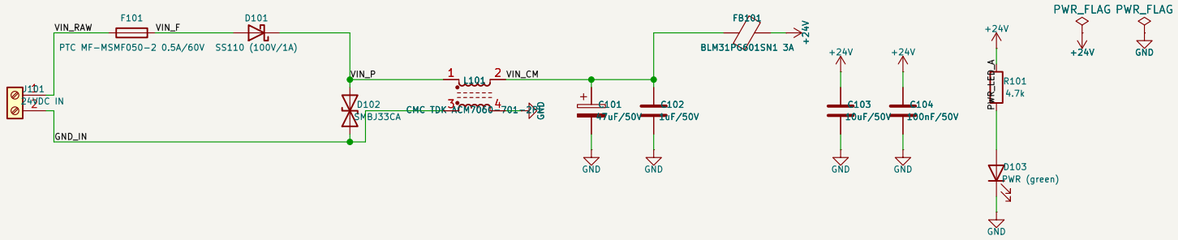
Sheet: Root
PLC STM32F407 + W5500 Modbus TCP

Sheet: /
File: PLC_STM32_Modbus.kicad_sch

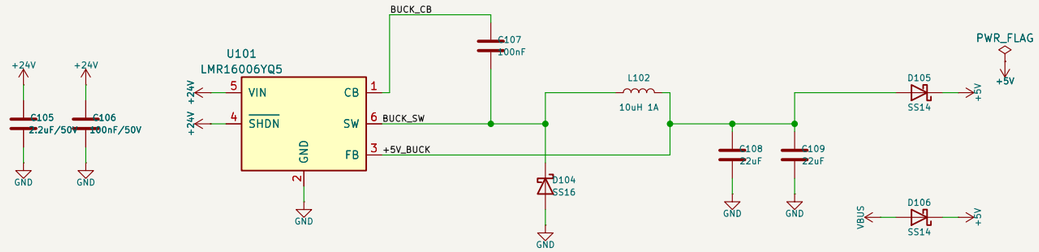
Title: PLC STM32F407 Modbus TCP – 16DO / 10AI

Size: A3	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/7

NGUON VAO 24VDC: cau chi -> chong nguoc -> TVS -> loc common-mode -> loc LC -> +24V

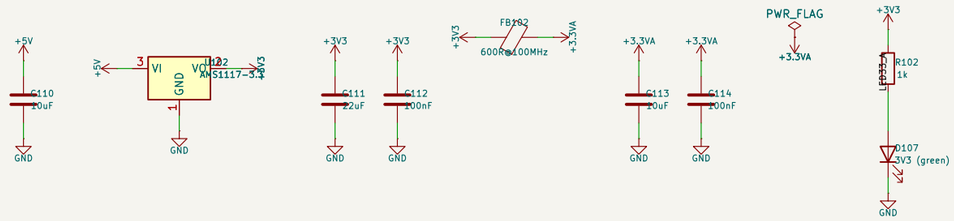


BUCK 24V -> 5V (LMR16006Y, 60V, 0.6A, 2.1MHz)



Cap nguon tu USB khi chua co 24V (de nap code)

LDO 5V -> 3.3V, loc rieng 3.3VA cho ADC



Sheet: Nguon 24VDC
PLC STM32F407 + W5500 Modbus TCP
Sheet: /Nguon 24VDC/
File: power.kicad_sch
Title: Nguon: loc 24VDC, buck 5V, LDO 3.3V, ref 3.0V
Size: A3 | Date: 2026-10-14 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 2/7

RESET / BOOT

NRST
C216
100nF
GND
SW201
NRST
RESET
GND
BOOT0
R201
10k
GND
+3V3
SW202
BOOT0
BOOT

VREF+ = 3.000V

+3.3VA
U201
REF3030
IN
GND
OUT
GND
HSE 8MHz
C214
1uF
C215
100nF
GND
OSCIN
C201
20pF
GND
OSC_OUT
Y201
8MHz
C202
20pF
GND
VREF_3V0

U200
STM32F407VET6
VBAT
VDD
VDD
VDD
VDD
VDD
VDD
VDD
VDDA
+3.3VA
NRST
BOOT0
VREF+
PH0
PH1
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
VCAP_1
VCAP_2
VSS
VSSA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15

DO2
DO1
DO16
DO15
DO14
DO13
DO12
DO11
DO10
DO9
ETH_RST
ETH_INT
AI1_ADC
AI2_ADC
AI3_ADC
AI4_ADC
UART1_TX
UART1_RX
SWCLK
SWDIO
SWO
BOOT1
DO8
DO7
DO6
DO5
DO4
DO3
ETH_CS
ETH_SCK
ETH_MISO
ETH_MOSI
AI5_ADC
AI6_ADC
AI7_ADC
AI8_ADC
AI9_ADC
AI10_ADC

LED trang thai

D201
RUN (green)
RUN_A
R203
1k
LED_RUN
D202
ERR (red)
ERR_A
R204
1k
LED_ERR

VCAP1
VCAP2
C213
2.2uF
C212
2.2uF

SWD debug (ST-Link)
+3V3
J201
SWD
VRef
NRST
RESET
SWCLK/TCK
SWDIO/TMS
SWO/TDO
NC/TDI
GND
GNDDetect
GND
GND

R202
10k
GND

GAN CHAN:

DO1..DO16 -> PE1,PE0,PB9,PB8,PB7,PB6,PB5,PB4,PD7..PD0

PA0-PA3, PC0-PC5 -> AI1..AI10 (ADC1 12-bit)

SPI2 PB12(CS) PB13 PB14 PB15 -> W5500, PD8 RST, PD9 INT

USART1 PA9/PA10 -> CH340C (bootloader UART)

PA13/PA14/PB3 -> SWD, PD12/PD13 -> LED

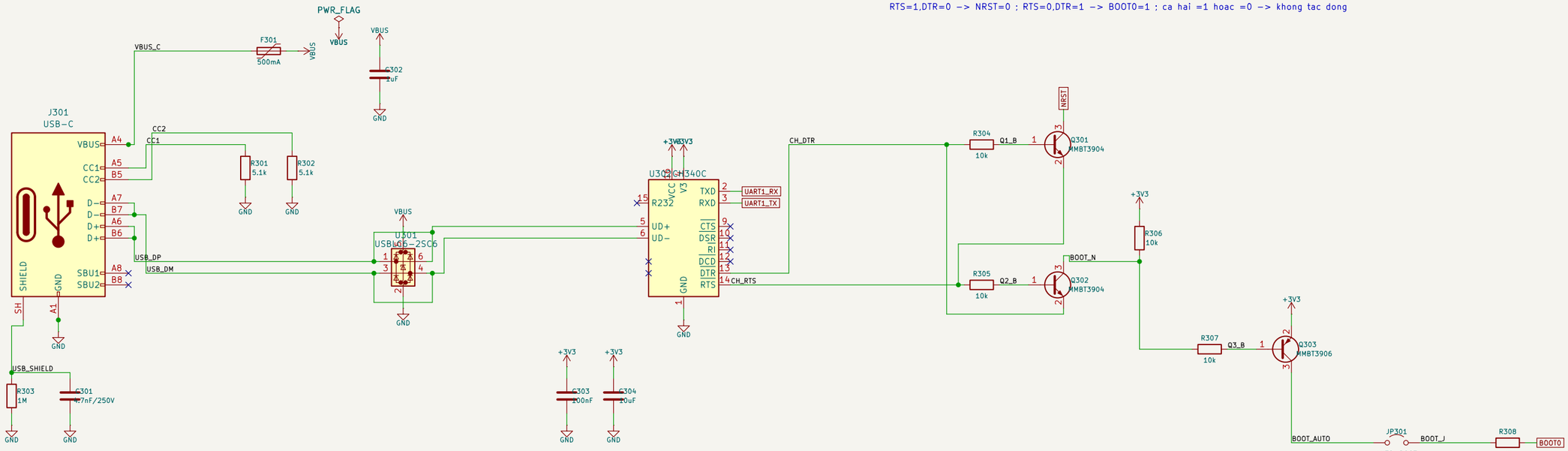
Tu loc nguon: 1 tu 100nF sat moi chan VDD; 1uF+100nF cho VDDA

+3V3
C203
100nF
C204
100nF
C205
100nF
C206
100nF
C207
100nF
C208
100nF
C209
4.7uF
+3.3VA
C210
1uF
C211
100nF
GND

Sheet: MCU STM32F407VET6
PLC STM32F407 + W5500 Modbus TCP
Sheet: /MCU STM32F407VET6/
File: mcu.kicad_sch
Title: Vi dieu khien STM32F407VET6
Size: A3 | Date: 2026-10-14 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 3/7

MACH TU DONG NAP (DTR/RTS) – giong ESP32

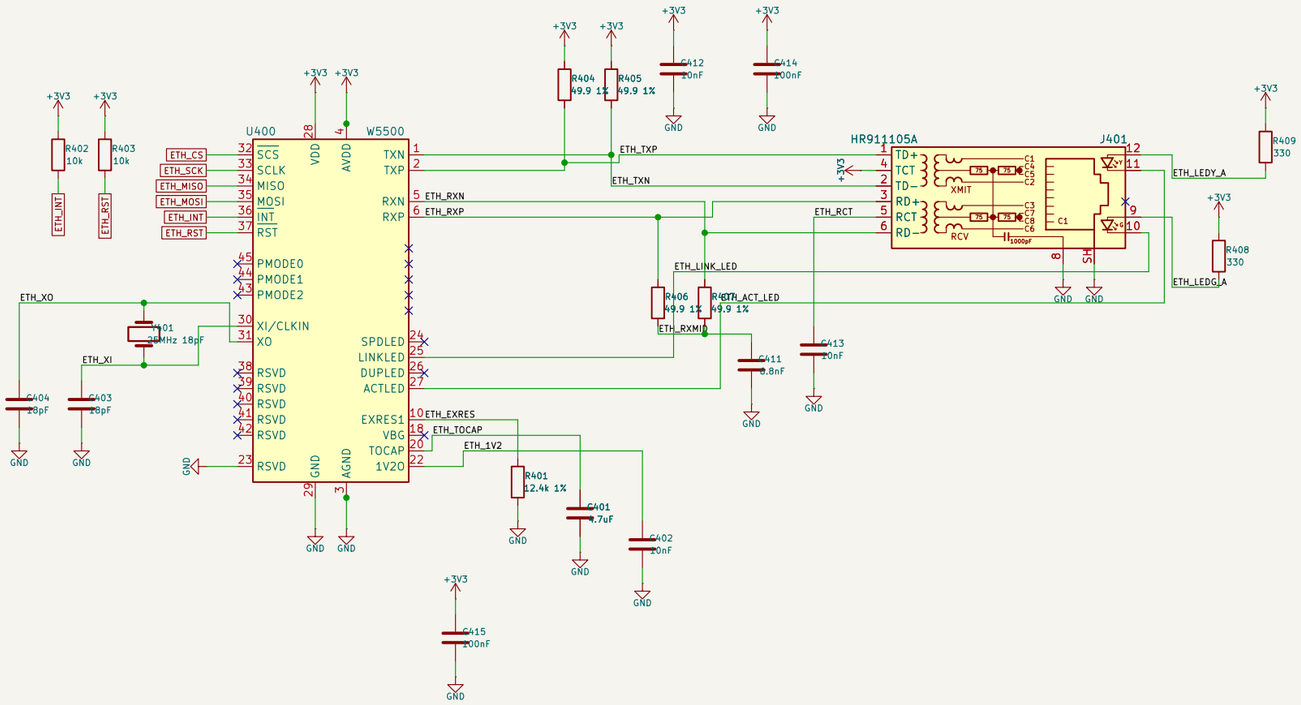
RTS=1,DTR=0 -> NRST=0 ; RTS=0,DTR=1 -> BOOT0=1 ; ca hai =1 hoac =0 -> khong tac dong



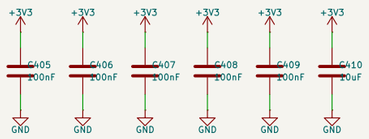
Sheet: USB nap code
PLC STM32F407 + W5500 Modbus TCP
Sheet: /USB nap code/
File: usb.kicad_sch
Title: USB–C -> CH340C -> USART1 + mach tu dong vao bootloader

Size: A3	Date: 2026–10–14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 4/7

Phoi hop tro khang PHY (theo WIZnet ref. schematic – kiem tra lai)



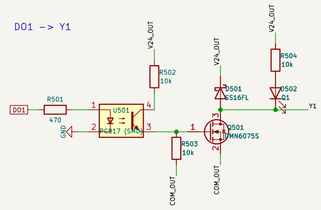
Tu loc W5500 (dat sat chan AVDD/VDD)



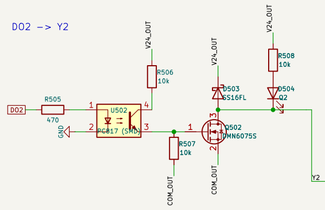
Sheet: Ethernet		
PLC STM32F407 + W5500 Modbus TCP		
Sheet: /Ethernet/		
File: comm.kicad_sch		
Title: W5500 Ethernet – Modbus TCP		
Size: A3	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 5/7

16 NGO RA TRANSISTOR SINK (NPN-type): MCU -> PC817 -> MOSFET 60V/2.5A. Tai noi giua V+ (24V) va Yn. Cap nguon ngoai vao V+/COM (cach ly voi mach logic).

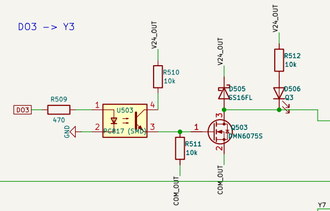
DO1 -> Y1



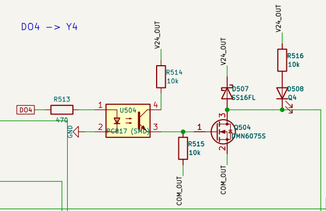
DO2 -> Y2



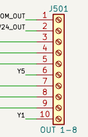
DO3 -> Y3

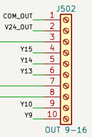


DO4 -> Y4

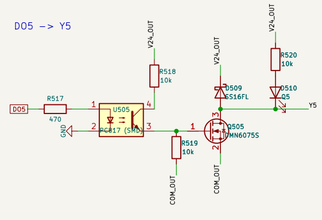


CAU DAU NGO RA

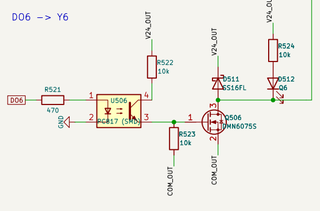




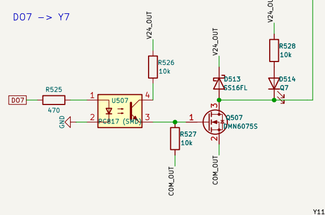
DO5 -> Y5



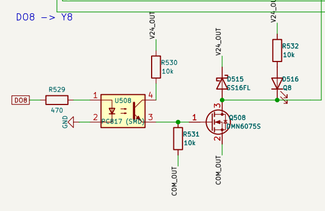
DO6 -> Y6



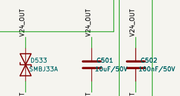
DO7 -> Y7



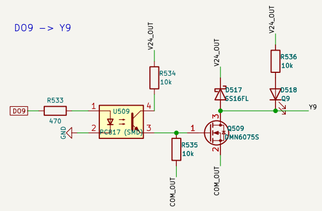
DO8 -> Y8



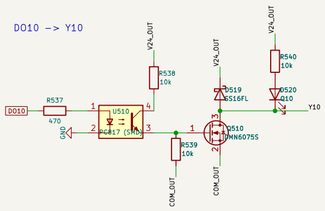
Loc / bao ve nguon ngo ra



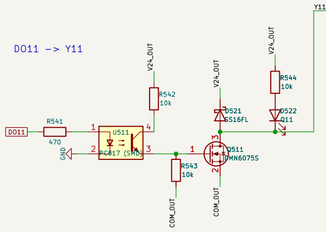
DO9 -> Y9



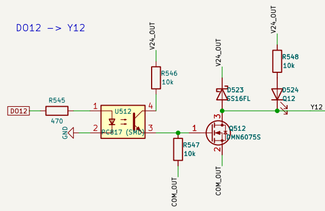
DO10 -> Y10



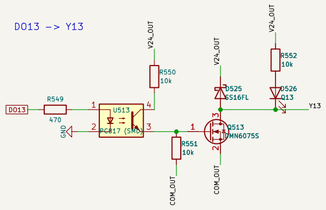
DO11 -> Y11



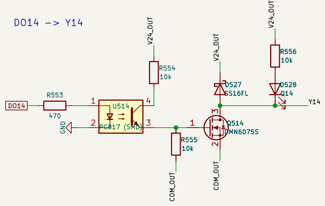
DO12 -> Y12



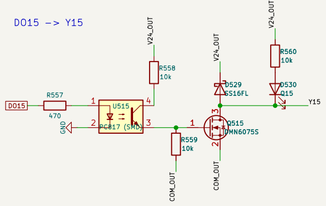
DO13 -> Y13



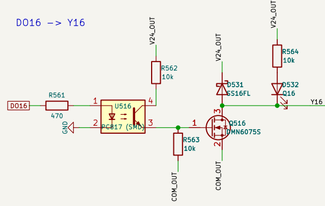
DO14 -> Y14



DO15 -> Y15



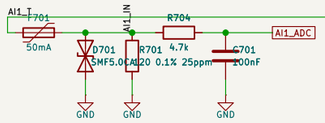
DO16 -> Y16



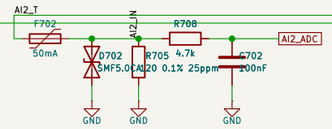
Sheet: 16 ngo ra transistor
PLC STM32F407 + W5500 Modbus TCP
Sheet: /16 ngo ra transistor/
File: outputs.kicad_sch
Title: 16 ngo ra transistor (sink), cach ly quang
Size: A2 | Date: 2026-10-14 | Rev: 1.0
KiCad E.D.A. 10.0.6 | Id: 6/7

10 NGO VAO 4–20mA: shunt 120R 0.1% –> Vadc = I x 120R ; 4mA –> 0.48V ; 20mA –> 2.40V ; 24mA –> 2.88V (VREF = 3.000V)

AI1: 4–20mA



AI2: 4–20mA

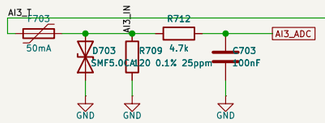


J701

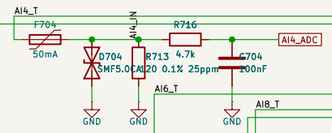


AI 1–10 (4–20mA)

AI3: 4–20mA



AI4: 4–20mA



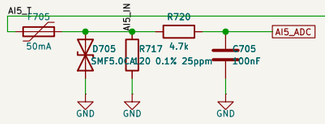
AI6_T

AI8_T

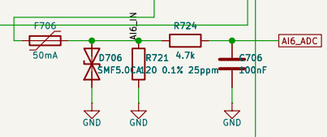
AI10_T

Nguon cap cho cam bien 2 day (24V loop) + chan chung GND

AI5: 4–20mA



AI6: 4–20mA



+24V

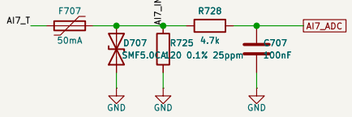


J703

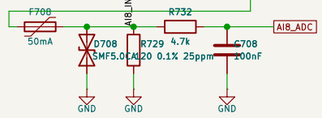


AI PWR/COM

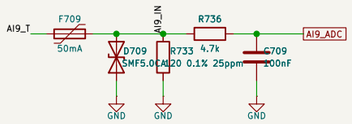
AI7: 4–20mA



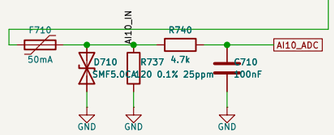
AI8: 4–20mA



AI9: 4–20mA



AI10: 4–20mA



Sheet: 10 ngo vao analog
PLC STM32F407 + W5500 Modbus TCP
Sheet: /10 ngo vao analog/
File: analog.kicad_sch
Title: 10 ngo vao analog 4–20mA

Size: A3	Date: 2026–10–14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 7/7